Noise and Vibration Technical Assessment

Post Mining Land Use – Rezoning - Macquarie Coal Complex Pilot Project

Department of Planning, Housing and Infrastructure

Prepared by:

SLR Consulting Australia Pty Ltd

SLR Project No.: 630.032981.00001

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Making Sustainability Happen

Revision Record

Revision	Date	Prepared By	Checked By	Authorised By
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Basis of Report

This report has been prepared by SLR Consulting Australia Pty Ltd (SLR) with all reasonable skill, care and diligence, and taking account of the timescale and resources allocated to it by agreement with Aurecon Australasia Pty Ltd (the Client). Information reported herein is based on the interpretation of data collected, which has been accepted in good faith as being accurate and valid. SLR may have used AI in the preparation of this document.

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SLR disclaims any responsibility to the Client and others in respect of any matters outside the agreed scope of the work.

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Acronyms and Abbreviations

AS	Australian Standard
AV:ATG	Assessing Vibration: a technical guideline (DEC, 2006)
BS	British Standard
dBA	A-weighted decibel (referenced 20 μPa)
DPHI	Department of Planning, Housing and Infrastructure
DECC	Department of Environment and Climate Change (now NSW EPA)
DIN	Deutches Institut für Normung (German Institute for Standardisation)
EPA	Environment Protection Authority
HNA	Highly Noise Affected
Hz	Hertz
ICNG	Interim Construction Noise Guideline (DECC, 2009)
ISO	International Standards Organisation
L_{Aeq}	Equivalent continuous noise level, providing a representation of the cumulative level of noise exposure over a defined period.
$L_{Aeq(15hour)}$	The equivalent continuous noise level for the 15-hour daytime period of 7.00 am to 10.00 pm
$L_{Aeq(9hour)}$	The equivalent continuous noise for the 9-hour night-time period of 10.00 pm to 7.00 am
$L_{Aeq(1hour)}$	The equivalent continuous noise for the 1-hour daytime or night-time period that has the potential to result in the greatest noise impact to sensitive receivers.
L_{Amax}	The maximum noise level during the measurement or assessment period. The L_{AFmax} or Fast is averaged over 0.125 of a second and the L_{ASmax} or Slow is averaged over 1-second.
NCA	Noise Catchment Area
NML	Noise Management Level
NSW	New South Wales
NPfI	Noise Policy for Industry (EPA, 2017)
RBL	Rating Background Level
RNP	Road Noise Policy (DECCW, 2011)
TfNSW	Transport for New South Wales
VDV	Vibration Dose Value

1.0 Introduction

SLR Consulting Australia Pty Ltd (SLR) has been engaged by Aurecon Pty Ltd (Aurecon) on behalf of the Department of Planning, Housing and Infrastructure (DPHI) to undertake a Noise and Vibration Technical Assessment (NVTA) to inform planning considerations associated with the proposed state-led rezoning of the Macquarie Coal Complex.

The Macquarie Coal Complex Pilot Project 2026 (the Master Plan) details the overall planning direction for the proposed rezoning to support coordinated transition from post mining land use to alternative employment generating land uses. This report outlines noise constraints and establishes the method for determining specific noise performance criteria for the structure plan by considering the proposed land uses and their impact on existing and future sensitive land uses and presents in-principal noise mitigation and management strategies to ensure the acoustic amenity of the local community and individual sensitive land uses surrounding the proposed rezoning are not adversely impacted.

SLR is a member of the Australian Acoustical Society (AAS) and a member firm of the Association of Australasian Acoustical Consultants (AAAC). The following report uses specialist acoustic terminology. An explanation of common terms is provided in **Appendix A**.

1.1 Precinct Description

The Macquarie Coal Complex Land Transformation Precinct forms part of the NSW Government's investigation into future land use pathways for mining affected land. The Master Plan represents a future-focused commitment to a transition for mining-dependent communities. It is complementary to the NSW Government's broader regional priorities and investment pipeline, as well as supporting Lake Macquarie City Council's emphasis on investment in long term public value for the community.

The Rezoning Proposal applies to 1,160 hectares of land and includes land at Macquarie Coal Complex (the Precinct) and Teralba Southgate sub-precinct, collectively referred to as the Site.

Macquarie Coal Complex

The Precinct sits adjacent to the Lake Macquarie Northwest Catalyst Area and is bounded by the suburb of Barnsley to the north, Cockle Creek and the Central-Coast Newcastle main passenger and freight train line to the east, Teralba and Wakefield to the south and Killingworth to the west. The M1 Pacific Motorway and Sugarloaf State Conservation Area are also located within in close proximity.

Teralba Southgate

Teralba Southgate sub-precinct site is located approximately 1.5km southeast of the main Macquarie Coal Complex site, 400m from Teralba Train Station and 250m from Booragul Train Station. Teralba Transport Oriented Development site (TOD) is adjacent to the north, and Booragul TOD adjoins the south of Teralba Southgate. Teralba Southgate is approximately 7 hectares and is comprised of existing warehouses, sheds and hardstand areas associated with historical mining uses of the site.

The Macquarie Coal Complex Land Transformation Precinct will deliver a dynamic, employment-led precinct which stimulates investment in the region and the creation of jobs. Leveraging strategic connections by road and rail, the Precinct will attract established and emerging industries to contribute to a diverse post-mining economy for the region. Protection and enhancement of important natural systems, cultural and historic assets, together with integrated Connecting with Country initiatives, will deliver long-term community and environmental benefit.

The Precinct and Teralba Southgate sub-precinct is located in a region with a strong and transferable skills base in mining, heavy industry, logistics, construction and land management. As regional mining operations transition toward closure, these skills represent a significant asset that can be redirected to support new and emerging industries.

The Structure Plan identifies the arrangement of developable areas, environmentally constrained areas, recreational areas, movement networks and key constraints across the Precinct and outlines the spatial logic for future development at a precinct scale, providing a framework within which detailed design, subdivision and development assessment will occur.

1.2 Purpose of this report

This NVTA has been prepared to consider the potential noise impact requirements applicable to the Precinct. The purpose of the report is to highlight the surrounding sensitive receivers, detail the methodology for establishing an applicable criteria and highlighting the potential land uses and the potential impact on existing, and future surrounding receivers.

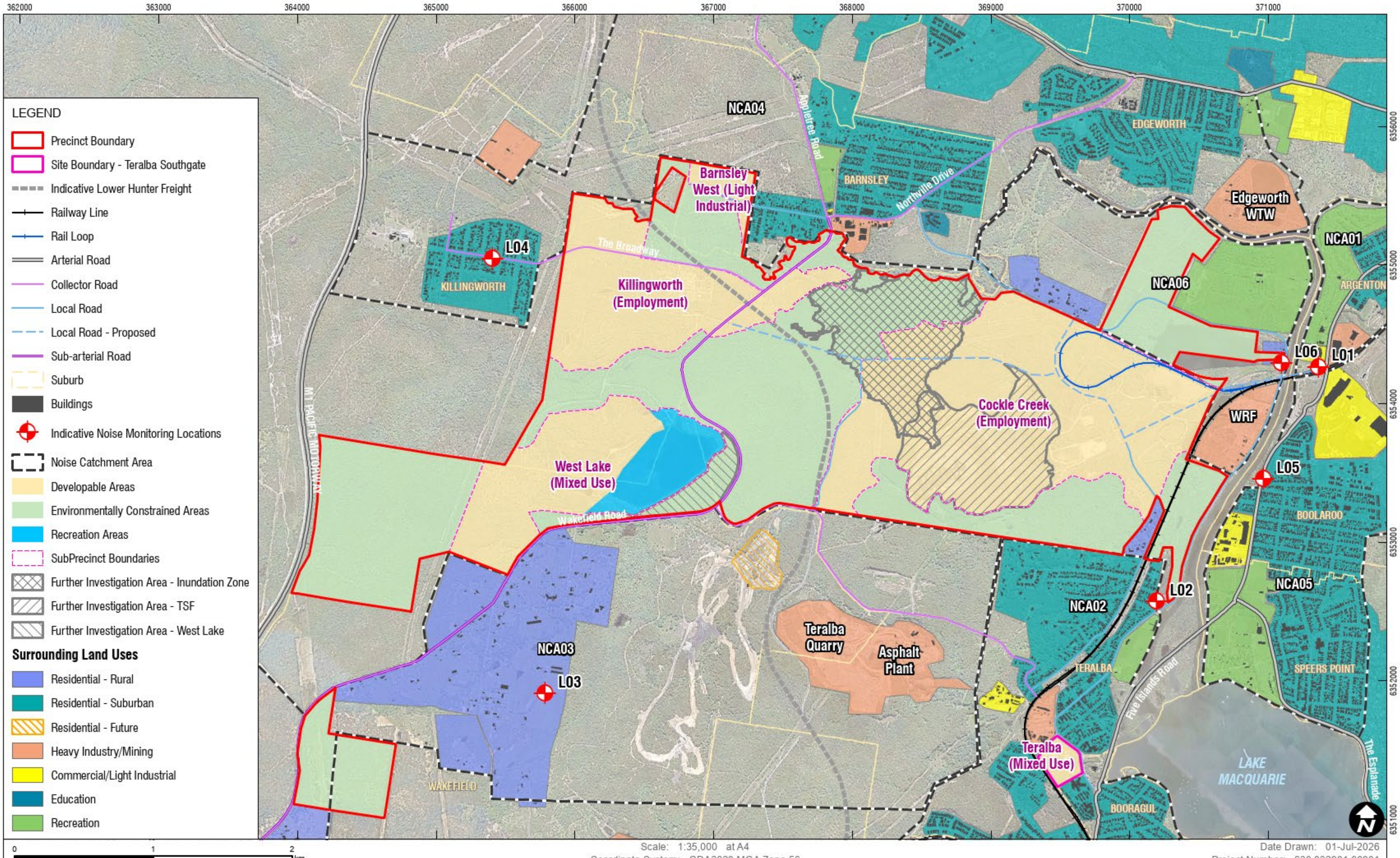
A detailed assessment will be undertaken of the Precinct once the nature and locations of the specific land uses are known as the Precinct begins to be developed.

1.3 Noise Sensitive Receivers Surrounding the Precinct

Sensitive receivers surrounding the Precinct have been identified and have been grouped into several Noise Catchment Areas (NCAs). The nearest receivers are shown in **Figure 1** and detailed in **Table 1**, with the Structure Plan presented in **Figure 2**.

Table 1 Surrounding Sensitive Receivers

NCA	Description	Receiver Type	Approximate Distance	Direction
NCA01	Residential Receivers along Lake Road	Residential Recreational	1,400m	North east
NCA02	Residential Receivers surrounding Teralba and Booragul station	Residential Educational	100m from the Precinct (25m from Teralba Southgate Sub-Precinct)	South east (surrounding Teralba Southgate Sub-Precinct)
NCA03	Residential Receivers along Wakefield Road	Residential Educational	70m from the Precinct	South
NCA04	Residential Receivers north of the Precinct	Residential Educational Recreational	25m from the Precinct	North
NCA05	Residential Receiver east of Cockle Creek.	Residential	500m from the Precinct	East
NCA06	Residential Receivers along The Weird Road	Residential Recreational	25m from the Precinct	North East
Commercial/ Industrial	Various commercial / industrial receivers surrounding the Precinct	Commercial/ industrial	Various	Various



Scale: 1:35,000 at A4
Coordinate System: GDA2020 MGA Zone 56
Data Source: NearMap May 2026

Date Drawn: 01-Jul-2026
Project Number: 630.032981.00001

SLR

SITE PLAN

FIGURE 1

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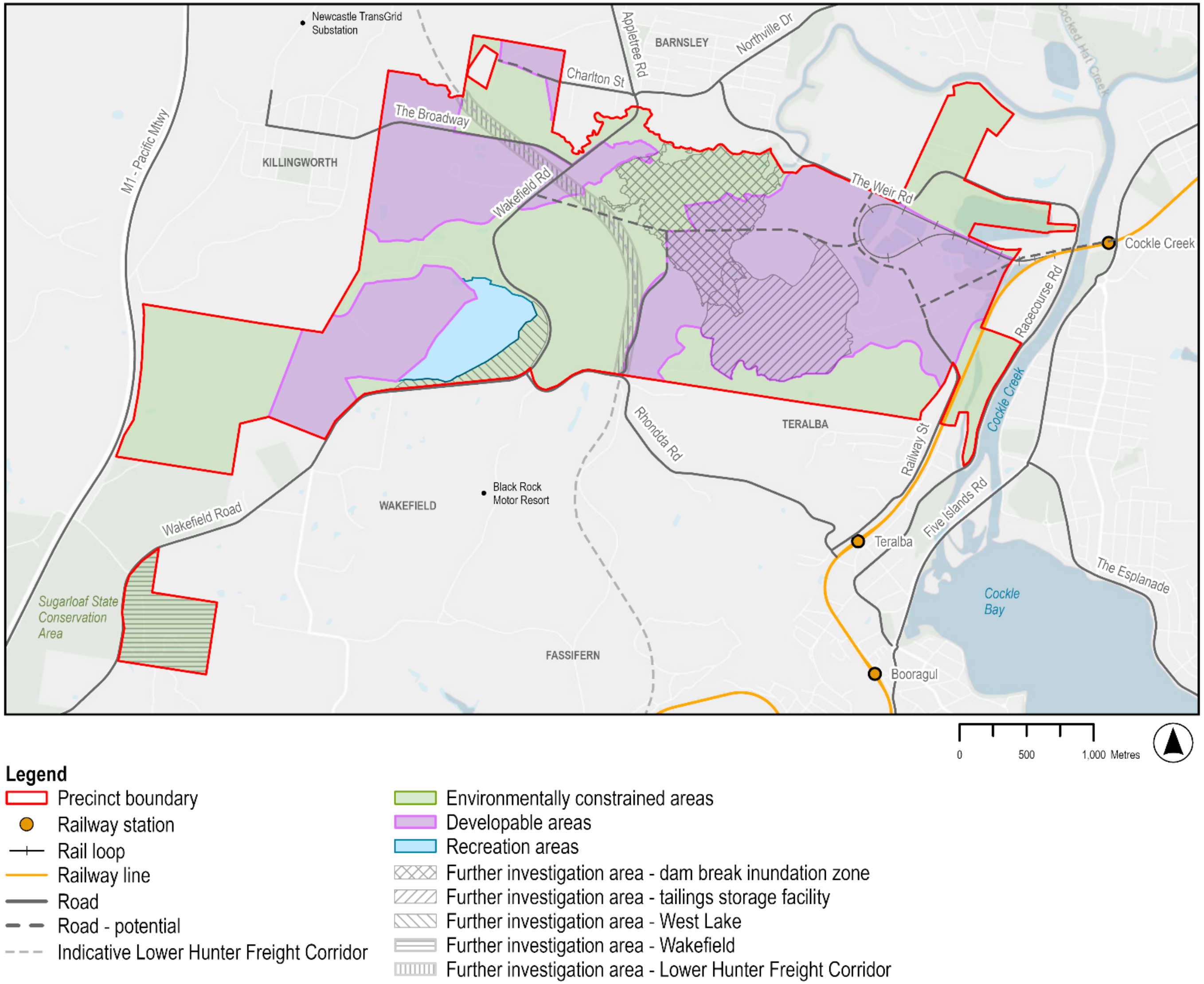
Figure 2 Structure Plan



2.0 Existing Noise Environment

2.1 Significant Noise Sources Surrounding the Precinct

The Precinct is situated near several major transport corridors and industrial operations, with a combination of traffic noise from arterial roads, railway lines and noise from surrounding industrial sites contributing to the existing ambient noise environment of the area.

2.1.1 Road Traffic Noise

The Precinct is bounded by major arterial roads including the M1 Pacific Motorway (M1) to the West and Five Islands Road to the East, carrying significant passenger and heavy vehicle volumes with Annual Average Daily Traffic (AADT) volumes of approximately 80,000 vehicles along the M1 and in excess of 22,000 vehicles along Five Islands Road.

Road traffic noise is expected to dominate the ambient noise environment for receivers close to these major road networks, particularly to the west and east of the Precinct.

2.1.2 Rail Noise

Noise from both passenger and freight rail operating on the Main North rail Line/Central Coast and Newcastle (CCN) Rail Corridor to the east of the Precinct also contributes to the existing ambient noise environment of the area, particularly for receivers to the east in Teralba, Boolaroo and other nearby suburbs.

An existing rail loop accesses the Precinct in the northeast of site, west of Cockle Creek Station.

The proposed Lower Hunter Freight Corridor (LHFC) alignment also runs through the west of the Precinct. The corridor alignment was protected by the NSW Government in 2022. The LHFC will provide for a dedicated freight rail line between Fassifern and Hexham, bypassing Newcastle's urban area. Separating freight and passenger rail operations aims to improve reliability and capacity on both freight and passenger rail networks.

Mitigation measures may be required to address the impact of rail noise where the railway is closest to proposed sub-precincts depending on the sensitivity of future developments to noise and/or vibration.

2.1.3 Industrial Noise

Existing mining operations to the south (Newstan Colliery and Downer Asphalt Plant) and heavy industrial premises to the east (Concrush) as well as other scattered industrial premises surrounding the Precinct (Holcim – Teralba Concrete and Structural Concrete Industries) would generate noise impacts to any sensitive developments such as residential usages.

Newstan Colliery is currently in Care and Maintenance with a State Significant Development Application lodged in 2020 to restart mining activities.

2.2 Existing Ambient Noise Environment

Existing publicly available ambient noise data has been obtained from noise monitoring performed for various projects at various locations around the Precinct to provide an indicative understanding of the existing noise environment within and surrounding the Precinct. The location of the noise monitoring locations used are shown in **Figure 1**

.

Table 2 Summary of Unattended Noise Monitoring Results from Surrounding Projects

ID	Year of Noise Monitoring	Address	Measured Noise Levels (dBA)					
			Background Noise (RBL)			Average Noise (L_{Aeq})		
			Day[1]	Evening[1]	Night[1]	Day[1]	Evening[1]	Night[1]
L01	2026	60 Lake Road, Argenton	44	41	38	55	54	51
L02	2025	2 Blair Street, Teralba	45	42	32	58	52	49
L03	2018	21 Jefferson Road, Wakefield	38	37	35	45	42	45
L04	2025	21 The Broadway, Killingworth	37	37	35	59	63	55
L05	2017	35 First Street, Teralba	49	42	32	67	67	62
L06	2017	14 Racecourse Road, Teralba	42	41	31	60	57	54

Note 1: The assessment periods are the daytime which is 7 am to 6 pm Monday to Saturday and 8 am to 6 pm on Sundays and public holidays, the evening which is 6 pm to 10 pm, and the night-time which is 10 pm to 7 am on Monday to Saturday and 10 pm to 8 am on Sunday and public holidays. See the NSW EPA Noise Policy for Industry.

As the project progresses, it is expected that updated noise monitoring will be undertaken surrounding the Precinct to build a complete picture of the existing noise environment.

3.0 Assessment Criteria

This section provides an overview of the main assessment criteria applicable to the proposed development.

- Operational Noise
    - NSW *Noise Policy for Industry* (NPfI) (EPA, 2017)
- Road Traffic Noise
    - *Road Noise Policy* (RNP) (DECCW, 2011)
- Noise Intrusion on new residential multiuse developments within the site
    - *Development near Rail Corridors and Busy Roads – Interim Guideline* (DOP, 2008)
    - *AS2107:2016 Acoustics – Recommended design sound levels and reverberation times for building interiors*
- Construction Noise
    - *Interim Construction Noise Guideline* (ICNG (DECC, 2009)
- Vibration
    - *Assessing Vibration: a technical guideline* (DEC,2006)
    - *BS 7385 Part 2 – 1993 Evaluation and measurement for vibration in buildings Part 2*, BSI, 1993
    - German Standard DIN 4150-3:1999-02 Structural vibration – Effects of vibration on structures (DIN 4150) in relation to vibration sensitive heritage structures.

3.1 Operational Noise Criteria

The NSW *Noise Policy for Industry* (NPfI) was released in 2017 and sets out the requirements for the assessment and management of operational noise from industry in NSW.

The NPfI defines how to determine 'trigger levels' for noise emissions from industrial developments. Where a development is likely to exceed the trigger levels at existing noise sensitive receivers, feasible and reasonable noise management measures are required to be considered to reduce the impacts.

There are two types of trigger levels – one to account for 'intrusive' noise impacts and one to protect the 'amenity' of particular land uses.

Intrusive and amenity noise levels are not used directly as regulatory limits. They are used to assess the potential impact of noise, assess feasible and reasonable mitigation options and subsequently determine achievable noise requirements.

3.1.1 Intrusive Criteria

The **intrusiveness** of an industrial noise source is generally considered acceptable if the L_{Aeq} noise level of the source, measured over a period of 15-minutes, does not exceed the representative background noise level by more than 5 dB. Intrusive noise levels are only applied to residential receivers. For other receiver types, only the amenity levels apply.

3.1.2 Amenity Criteria

To limit continual increases in noise levels from the use of the intrusiveness level alone, the ambient noise level within an area from all industrial sources should remain below the recommended amenity levels specified in the NPfI for that particular land use as shown in **Table 3**.

The NPfI provides guidance on assigning residential receiver amenity noise categories based on the site-specific features shown in **Table 4**.

Table 3 Amenity Noise Levels

Receiver	Noise Amenity Area	Time of Day	Recommended Amenity Noise Level L_{Aeq} (dBA)
Residential	Rural	Day	50
		Evening	45
		Night	40
	Suburban	Day	55
		Evening	45
		Night	40
	Urban	Day	60
		Evening	50
		Night	45
Hotel, Motels, Holiday accommodation			5dBA above the recommended amity noise level for residences for the relevant noise amenity area
School classroom - Internal	All	Noisiest 1 hour period when in use	35
Place of Worship – Internal	All	When in use	40
Area specifically reserved for passive recreation (eg national park)	All	When in use	50
Active recreation (school playground, golf course)	All	When in use	55
Commercial premises	All	When in use	65
Industrial premises	All	When in use	70

Table 4 Residential Receiver Amenity

Receiver Category	Typical Planning Land Use Zoning	Typical Existing Background Noise Levels (RBL)	Description
Rural	RU1 – primary production RU2 – rural landscape RU4 – primary production small lots R5 – large lot residential E4 – environmental living	Daytime <40 dBA Evening <35 dBA Night <30 dBA	Rural – an area with an acoustical environment that is dominated by natural sounds, having little or no road traffic noise and generally characterised by low background noise levels. Settlement patterns would be typically sparse. Note: Where background noise levels are higher than those presented due to existing industry or intensive agricultural activities, the selection of a higher noise amenity area should be considered.
Suburban residential	RU5 – village RU6 – transition R2 – low density residential R3 – medium density residential E2 – environmental conservation E3 – environmental management	Daytime <45 dBA Evening <40 dBA Night <35dBA	Suburban – an area that has local traffic with characteristically intermittent traffic flows or with some limited commerce or industry. This area often has the following characteristic: evening ambient noise levels defined by the natural environment and human activity.
Urban residential	R1 – general residential R4 – high density residential B1 – neighbourhood centre (boarding houses and shop-top housing) B2 – local centre (boarding houses) B4 – mixed use	Daytime >45 dBA Evening >40 dBA Night >35 dBA	Urban – an area with an acoustical environment that: • Is dominated by ‘urban hum’ or industrial source noise, where urban hum means the aggregate sound of many unidentifiable, mostly traffic and/or industrial related sound sources • Has through-traffic with characteristically heavy and continuous traffic flows during peak periods • Is near commercial districts or industrial districts • Has any combination of the above.

A screening assessment to determine the suitable amenity noise categories for the surrounding receivers based on the previous noise monitoring have been determined with reference to the applicable land uses surrounding the Precinct.

A summary of preliminary outcomes is shown in **Table 5**.

Table 5 Residential Receiver Amenity Screening Assessment

NCA	Land Use Zoning	Existing Background Noise Levels RBL (dBA)			Indicative Amenity Classification
		Day	Eve	Night	
NCA01	R2 – Low density residential R3 – Medium density residential	44	41	38	Suburban
NCA02	R2 – Low density residential R3 – Medium density residential E1 – Local Centre	45	42	32	Suburban
NCA03	RU2 – Rural Landscape	38	37	35	Rural
NCA04	R2 – Low density residential R3 – Medium density residential RU6 – Transition	37	37	35	Suburban
NCA05	R2 – Low density residential R3 – Medium density residential	49	42	32	Suburban
NCA06	RU4 – Primary Production small lots	42	41	31	Rural

The recommended amenity levels presented in **Table 3** represent the objective for total industrial noise at a receiver location, whereas the project amenity noise level represent the objective for noise from a single industrial development

To ensure that industrial noise levels (existing plus new) remain within the recommended amenity noise levels for an area, a project amenity noise level applies for each new source of industrial noise as follows:

Project amenity noise levels for industrial developments = recommended amenity noise level minus 5dB

The project amenity noise level nominated here therefore applies to the whole Precinct development rather than individual sites within the Precinct.

3.1.2.1 Areas of high traffic noise

The level of transport noise may be high enough to make noise from an industrial source effectively inaudible, even though the L_{Aeq} noise level from that industrial noise source may exceed the project amenity noise level. In such cases, the project amenity noise level may be derived as follows:

High traffic project amenity noise level for industrial developments = $L_{Aeq(period)}$ minus 5dB

3.1.3 Project Noise Trigger Levels

The Project Noise Trigger Levels (PNTL) are the most stringent of the intrusiveness and amenity trigger level for each period.

Applicable PNTL will be established through detailed noise monitoring at the potentially most affected sensitive receivers around the Precinct as part of the planning approvals process.

3.1.4 Cumulative Noise Impacts

The NPfI aims to limit continuing increases in industrial noise through the application of amenity noise levels. The project amenity noise levels are reduced by 5 dB relative to the recommended amenity levels to allow for other sources of industrial noise in the area (existing and new). By doing this, the policy accounts for potential cumulative impacts by lowering the criteria for each individual development. The NPfI states that *"where the project amenity noise level applies and it can be met, no additional consideration of cumulative industrial noise is required."*

3.1.5 Sleep Disturbance

The potential for sleep disturbance from maximum noise level events during the night-time period from the proposal is required to be considered. This is applicable only to residential receivers.

The NPfI defines the sleep disturbance screening level as:

- $L_{Aeq(15minute)}$ 40 dB(A) or the prevailing RBL plus 5 dB, whichever is the greater, and/or
- L_{AFmax} 52 dB(A) or the prevailing RBL plus 15 dB, whichever is greater,

Applicable sleep disturbance screening criteria will be established through detailed monitoring as outlined in **Section 3.1.3**.

A detailed maximum noise level event assessment should be completed where the sleep disturbance screening level is exceeded. The detailed assessment should cover the maximum noise level, the extent to which the maximum noise level exceeds the RBL, and the number of times this happens during the night-time period.

The NPfI refers to the Road Noise Policy (RNP) for additional information regarding sleep disturbance. enHealth Council studies are referenced which indicate that for short-term or transient noise events, for good sleep over eight hours the indoor L_{AFmax} sound pressure level should ideally not exceed around 45 dBA more than 10 or 15 times per night.

The RNP goes on to conclude that from the research on sleep disturbance to date:

- Maximum internal noise levels below 50 dBA to 55 dBA are unlikely to awaken people from sleep.
- One or two events per night with maximum internal noise levels of 65-70 dBA are not likely to affect health and wellbeing significantly.

3.2 Rail Noise from Non-network Rail Lines on or Exclusively Servicing Industrial Sites

Rail related activities (such as movement of rolling stock on rail loops or sidings, loading and shunting activities etc) occurring within the boundary of an industrial premises as defined in an environment protection licence are to be assessed as part of the industrial premises using the *NSW Industrial noise policy* (EPA, 2000) (INP).

Where a non-network rail line exclusively servicing one or more industrial sites extends beyond the boundary of the industrial premises, noise from this section of track should be assessed against the recommended acceptable L_{Aeq} noise levels from industrial noise sources for the indicative noise amenity area as outlined in Table 2.1 of the NSW Industrial Noise Policy, reproduced in **Table 6**.

Table 6 Recommended L_{Aeq} Noise Levels from no-network Rail Lines

Receiver	Noise Amenity Area	Time of Day	L_{Aeq} Noise Level (dBA)	
			Acceptable	Recommended Maximum
Residential	Rural	Day	50	55
		Evening	45	50
		Night	40	45
	Suburban	Day	55	60
		Evening	45	50
		Night	40	45
	Urban	Day	60	65
		Evening	50	55
		Night	45	50
	Urban/Industrial Interface – for existing situations	Day	65	70
		Evening	55	60
		Night	50	55
School classroom - Internal	All	Noisiest 1 hour period when in use	35	40
Hospital ward - Internal	All	Noisiest 1 hour period when in use	35	40
Hospital ward - External			50	55
Place of Worship – Internal	All	When in use	40	45
Area specifically reserved for passive recreation (eg national park)	All	When in use	50	55
Active recreation (school playground, golf course)	All	When in use	55	60
Commercial premises	All	When in use	65	70
Industrial premises	All	When in use	70	75

3.3 Traffic on Surrounding Roads

The potential impacts from project related traffic on the surrounding public roads are assessed using the NSW EPA *Road Noise Policy* (RNP).

An initial screening test is first applied to evaluate if existing road traffic noise levels are expected to increase by more than 2.0 dB. Where this is considered likely, further assessment is required using the RNP criteria shown in **Table 7**.

Table 7 RNP / NCG Criteria for Assessing Traffic on Public Roads

Road Category	Type of Project/Land Use	Assessment Criteria (dBA)	
		Daytime (7 am – 10 pm)	Night-time (10 pm – 7 am)
Freeway/ arterial/ sub-arterial roads	Existing residences affected by additional traffic on existing freeways/arterial/sub-arterial roads generated by land use developments	$L_{Aeq(15hour)}$ 60 (external)	$L_{Aeq(9hour)}$ 55 (external)
Local roads	Existing residences affected by additional traffic on existing local roads generated by land use developments	$L_{Aeq(1hour)}$ 55 (external)	$L_{Aeq(1hour)}$ 50 (external)

3.4 Noise Intrusion

3.4.1 Criteria for Development Adjacent to Existing Road and Rail Infrastructure

The NSW Government's State Environmental Planning Policy (Transport and Infrastructure) 2021 (the SEPP) was introduced to aid the delivery of infrastructure across the State by improving regulatory certainty and efficiency.

In accordance with the SEPP, Table 3.1 of the NSW Department of Planning and Infrastructure's guideline *Development near Rail Corridors and Busy Roads – Interim Guideline* (DP&I, 2008), provides internal noise criteria for residential and non-residential buildings impacted by existing road and rail infrastructure. These criteria are summarised in **Table 8**.

Table 8 DP&I Interim Guideline Noise Criteria

Residential Buildings			
Type of Occupancy		Noise Level (dBA)	Applicable Time Period
Sleeping Areas (Bedrooms)		35	Night 10pm to 7am
Other habitable room (excl. garages, kitchens, bathrooms & hallways)		40	At any time
Non-Residential Buildings			
Type of Occupancy			**Recommended Max Noise Level (dBA)**
Educational Institutions including child care centres			40
Places of Worship			40
Hospitals	Wards		35
	Other Noise Sensitive Areas		45

Note 1: Airborne noise is calculated as $L_{Aeq(15hour)}$ daytime and $L_{Aeq(9hour)}$ night-time.

If internal noise levels with windows or doors open exceed the above criteria by more than 10 dB, then a natural ventilation path from a non-noise affected facade or forced ventilation system for the habitable rooms may be necessary to enable residents to leave windows closed during noisy periods.

Where windows must be kept closed, the adopted ventilation systems must meet the requirements of the National Construction Code Series Building Code of Australia, and Australian Standard 1668 – The use of ventilation and air conditioning in buildings.

It is generally accepted that internal noise levels in a dwelling are 10 dB lower than external noise levels with the windows open, and 20 dB lower than external noise levels with the windows closed and standard glazing. Where a traffic noise reduction of more than 20 dB is required, additional façade mitigation measures, such as thicker glazing and/or façade upgrades may be required.

3.4.2 AS2107 Acoustics – Recommended design sound levels and reverberation times for building interiors

AS2107:2016 details applicable internal noise criteria for difference spaces depending on their intended use. Where a use that is not detailed in the *Development near Rail Corridors and Busy Roads – Interim Guideline* as summarised in **Section 3.4.1**, reference can be made to AS2107:2016 to adopt an appropriate criteria for future developments within the Precinct.

3.5 Construction Noise

3.5.1 Interim Construction Noise Guideline

The NSW *Interim Construction Noise Guideline* (ICNG) is used to assess and manage impacts from construction noise on residences and other sensitive land uses in NSW.

The ICNG contains procedures for determining project specific Noise Management Levels (NMLs) for sensitive receivers based on the existing background noise in the area. The 'worst-case' noise levels from construction of a project are predicted and then compared to the NMLs in a 15-minute assessment period to determine the likely impact from construction activities associated with the Precinct.

The NMLs are not mandatory limits, however, where construction noise levels are predicted or measured to be above the NMLs, feasible and reasonable work practices to minimise noise emissions are to be investigated.

3.5.1.1 Residential Receivers

The ICNG approach for determining NMLs at residential receivers is shown in **Table 9**.

Table 9 ICNG NMLs for Residential Receivers

Time of Day	NML $L_{Aeq(15minute)}$	How to Apply
Standard Construction Hours Monday to Friday 7am to 6pm Saturday 8am to 1pm	Noise affected RBL[1] + 10 dB	The noise affected level represents the point above which there may be some community reaction to noise Where the predicted or measured $L_{Aeq(15minute)}$ is greater than the noise affected level, the proponent should apply all feasible and reasonable work practices to meet the noise affected level The proponent should also inform all potentially impacted residents of the nature of works to be carried out, the expected noise levels and duration, as well as contact details.

Time of Day	NML $L_{Aeq(15minute)}$	How to Apply
No work on Sundays or public holidays	Highly Noise Affected 75 dBA	The Highly Noise Affected (HNA) level represents the point above which there may be strong community reaction to noise Where noise is above this level, the relevant authority (consent, determining or regulatory) may require respite periods by restructuring the hours that the very noisy activities can occur, taking into account: Times identified by the community when they are less sensitive to noise (such as before and after school for works near schools or mid-morning or mid-afternoon for works near residences If the community is prepared to accept a longer period of construction in exchange for restrictions on construction times.
Outside Standard Construction Hours	Noise affected RBL + 5 dB	A strong justification would typically be required for works outside the recommended standard hours The proponent should apply all feasible and reasonable work practices to meet the noise affected level Where all feasible and reasonable practises have been applied and noise is more than 5 dB above the noise affected level, the proponent should negotiate with the community.

Note 1: The RBL is the Rating Background Level and the ICNG refers to the calculation procedures in the NSW *Industrial Noise Policy* (INP). The INP has been superseded by the NSW *Noise Policy for Industry* (NPfI).

3.5.1.2 Sleep Disturbance

The ICNG lists five categories of work that might need to be undertaken outside of Standard Construction Hours:

- the **delivery of oversized equipment or structures** that require special arrangements to transport on public roads
- **emergency work** to avoid the loss of life or damage to property, or to prevent environmental harm
- **maintenance and repair of public infrastructure** where disruption to essential services or considerations of worker safety do not allow work within standard hours
- **public infrastructure work** that shortens the length of the project and is supported by the affected community
- work where a proponent demonstrates and justifies **a need to operate outside the recommended standard hours**.

Where construction work is planned to extend over more than two consecutive nights, the ICNG recommends that an assessment of sleep disturbance impacts should be completed. The ICNG refers to the *NSW Environmental Criteria for Road Traffic Noise* for assessing the potential impacts, which notes that to limit the level of sleep disturbance, the L_1 level (or L_{Amax}) should not exceed the existing L90 background noise level by more than 15 dB.

3.5.1.3 'Other Sensitive' Land Uses

Several non-residential land uses have been identified in the study area. The NMLs for 'other sensitive' receivers are shown in **Table 10**.

Table 10 Construction NMLs for 'Other Sensitive' Land Uses

Land Use	Noise Management Level $L_{Aeq(15minute)}$ (dBA) (applied when the property is in use)	
	Internal	External
ICNG 'Other Sensitive' Receivers		
Classrooms at schools and other educational institutions	45	55[1]
Hospital wards and operating theatres	45	65[2]
Places of worship	45	55[1]
Active recreation areas (characterised by sporting activities which generate noise)	-	65
Passive recreation areas (characterised by contemplative activities that generate little noise)	-	60
Commercial	-	70
Industrial	-	75
Non-ICNG 'Other Sensitive' Receivers		
Child care centres – sleeping areas[3]	40	50[1]
Aged Care	Considered as Residential	

Note 1: It is assumed that these receivers have windows partially open for ventilation which results in internal noise levels being around 10 dB lower than the external noise level.

Note 2: It is assumed that these receivers have fixed windows which conservatively results in internal noise levels being around 20 dB lower than the external noise level.

Note 3: Criteria taken from Association of Australian Acoustical Consultants *Guideline for Child Care Centre Acoustic Assessment*.

3.6 Construction Vibration

The effects of vibration from construction works can be divided into three categories:

- Those in which the occupants of buildings are disturbed (human comfort)
- Those where building contents may be affected (building contents)
- Those where the integrity of the building may be compromised (structural or cosmetic damage).

3.6.1 Human Comfort Vibration

People can sometimes perceive vibration impacts when vibration generating construction works are located close to occupied buildings.

Vibration from construction works tends to be intermittent in nature and the EPA's *Assessing Vibration: a technical guideline* (2006) provides criteria for intermittent vibration based on the Vibration Dose Value (VDV). The 'preferred' and 'maximum' VDVs for human comfort impacts are shown in **Table 11**.

Table 11 Vibration Dose Values for Intermittent Vibration

Building Type	Assessment Period	Vibration Dose Value[1] ($m/s^{1.75}$)	
		Preferred	Maximum
Critical Working Areas (eg operating theatres or laboratories)	Day or night-time	0.10	0.20
Residential	Daytime	0.20	0.40
	Night-time	0.13	0.26
Offices, schools, educational institutions and places of worship	Day or night-time	0.40	0.80
Workshops	Day or night-time	0.80	1.60

Note 1: The VDV accumulates vibration energy over the daytime and night-time assessment periods and is dependent on the level of vibration as well as the duration.

3.6.2 Effects on Building Contents

People perceive vibration at levels well below those likely to cause damage to building contents. For most receivers, the human comfort vibration criteria are the most stringent and it is generally not necessary to set separate criteria for vibration effects on typical building contents.

Exceptions to this can occur when vibration sensitive equipment, such as electron microscopes, are located in buildings near to construction works. No such items of equipment have been identified in the proposal area.

3.6.3 Structural and Cosmetic Damage Vibration

If vibration from construction works is sufficiently high it can cause damage to structural elements of affected buildings. The levels of vibration required to cause cosmetic damage tend to be at least an order of magnitude (10 times) higher than those at which people can perceive vibration.

Examples of damage that can occur includes cracks or loosening of drywall surfaces, cracks in supporting columns and loosening of joints. Structural damage vibration limits are contained in British Standard BS 7385 and German Standard DIN 4150.

BS 7385

British Standard BS 7385 recommends vibration limits for transient vibration judged to give a minimal risk of vibration induced damage to affected buildings. The limits for residential and industrial buildings are shown in **Table 12**.

Table 12 BS 7385 Transient Vibration Values for Minimal Risk of Damage

Group	Type of Building	Peak Component Particle Velocity in Frequency Range of Predominant Pulse	
		4 Hz to 15 Hz	15 Hz and Above
1	Reinforced or framed structures. Industrial and heavy commercial buildings	50 mm/s at 4 Hz and above	
2	Unreinforced or light framed structures. Residential or light commercial type buildings	15 mm/s at 4 Hz increasing to 20 mm/s at 15 Hz	20 mm/s at 15 Hz increasing to 50 mm/s at 40 Hz and above

Note 1: Where the dynamic loading caused by continuous vibration may give rise to dynamic magnification due to resonance, especially at the lower frequencies where lower guide values apply, then the guide values may need to be reduced by up to 50%.

For heritage buildings, the standard states that *"a building of historical value should not (unless it is structurally unsound) be assumed to be more sensitive".*

DIN 4150

Heritage buildings and structures should be considered on a case-by-case basis but as noted in BS 7385, they should not be assumed to be more sensitive to vibration, unless structurally unsound. Where the heritage building or structure is structurally sound the vibration guide values for cosmetic damage in in **Table 12** would apply. Where a heritage building is deemed to be sensitive, the more stringent German Standard DIN 4150 Group 3 guideline values in **Table 13** can be applied.

Table 13 DIN 4150 Guideline Values for Short-term Vibration on Structures

Group	Type of Structure	Guideline Values Vibration Velocity (mm/s)				
		Foundation, All Directions at a Frequency of			Topmost Floor, Horizontal	Floor Slabs, Vertical
		1 to 10Hz	10 to 50Hz	50 to 100Hz	All frequencies	
3	Structures that, because of their particular sensitivity to vibration, cannot be classified as Group 1 or 2 and are of great intrinsic value (e.g. heritage listed buildings)	3	3 to 8	8 to 10	8	20

4.0 Screening Analysis

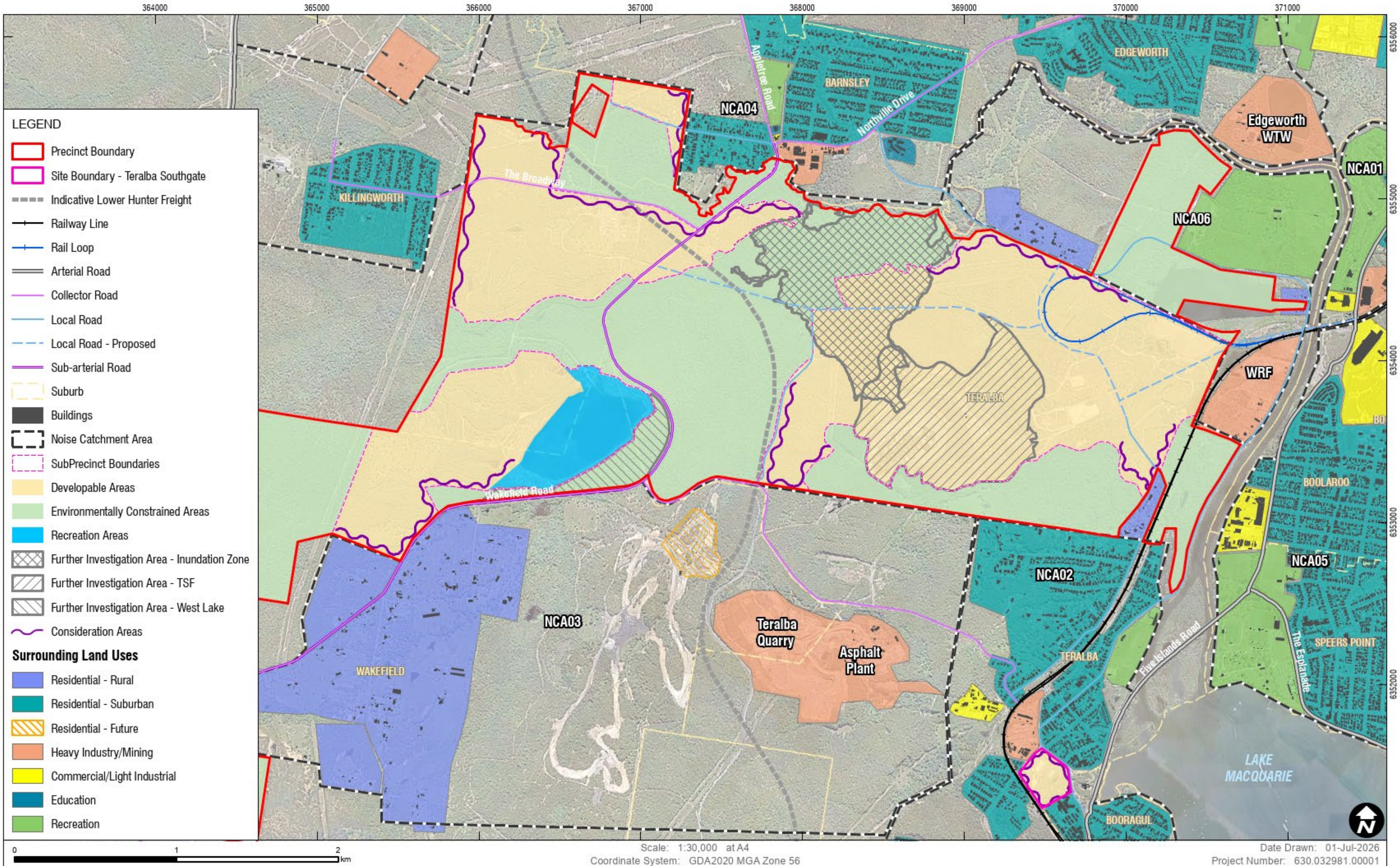
Noise considerations for the sub-precincts identified in **Figure 1** are summarised in **Table 14**, based on potential land use typology for each sub-precinct developed as part of the master plan.

Table 14 Noise Considerations for Proposed Sub-Precincts

Sub-Precinct	Description	Proposed Land Use Typology	Noise Considerations
1 – Cockle Creek	Employment	Circular Economy, Waste and Recycling; Industrial and Advanced Manufacturing; Data Centres and Digital Infrastructure; Freight Logistics, Distribution and Supply Chain Hub; Energy Transition & Renewable Infrastructure	• Industrial developments would be required to be assessed in accordance with the NPfI. Residential receivers in NCA06 are in close proximity to northern boundary of the Precinct. Land use planning should consider proximity of these sensitive receivers and consider buffer zones and design to incorporate appropriate noise mitigation measures including the judicious placement of future noise generating developments. Detailed assessment of potential future land uses should be undertaken to understand potential impact from developments in the sub-precinct on these receivers. • Energy Transition and Renewable Infrastructure (and in some cases industrial developments) have the potential to exhibit tonal and/or low frequency characteristics which can cause increased annoyance and sensitivity to nearby receivers. These characteristics must be considered in accordance with Fact Sheet C of the NPfI. • Any development which proposes utilising the existing rail spur would need to consider rail noise impacts associated with its use and should be assessed under the NPfI and the RING. Potential for curve noise due to tight radius of spur and may require consideration of specific mitigation measures. Detailed assessment would be required if proposed.
2 – Killingworth	Employment	Circular Economy, Waste and Recycling; Industrial and Advanced Manufacturing; Data Centres and Digital Infrastructure; Freight Logistics, Distribution and Supply Chain Hub; Energy Transition & Renewable Infrastructure	• Industrial developments would be required to be assessed in accordance with the NPfI. Residential receivers in NCA04 in close proximity to northern boundary. Land use planning should consider proximity of these sensitive receivers and consider buffer zones and design of precinct to develop appropriate noise mitigation measures. • Energy Transition and Renewable Infrastructure has the potential to exhibit tonal and/or low frequency characteristics which can cause increased annoyance and sensitivity to nearby receivers. These characteristics must be considered in accordance with Fact Sheet C of the NPfI. Night-time amenity levels and proximity to existing residential receivers in Killingworth would require detailed assessment to understand potential impact from 24-hour developments.

Sub-Precinct	Description	Proposed Land Use Typology	Noise Considerations
3 – West Lake	Mixed Use	Conservation, passive and active recreation, nature-based tourism and complementary low-impact uses.	• Development in this area would need to consider interface between any light industrial/commercial usage and potential recreation/tourism uses to ensure acoustic amenity is appropriate for those land uses. • Proximity to rural residential land uses south of Wakefield Rd and future luxury residential within the BlackRock Motor Resort, would also require consideration, particularly if night-time operations are proposed.
4 – Teralba Southgate	Mixed Use	Medium density residential, commercial. Community uses	• Sub-precinct located within 20m of existing freight and passenger rail line. Noise impacts would be assessed under the Development near roads and rail interim guideline for residential uses within assessment distance and would likely require consideration of good acoustic building design and construction principles to achieve internal noise levels. • Consideration of operational impacts from commercial developments on adjacent residential/aged care receivers should be assessed in accordance with the NPfI.
5 – Barnsley West	Light Industry	Light Industrial, Commercial	• Sub-Precinct located immediately west of residential pocket of Barnsley in NCA04. Industrial and commercial developments would be required to be assessed in accordance with the NPfI. Land use planning should consider proximity of these sensitive receivers and consider buffer zones and design of the sub-precinct to develop appropriate noise mitigation measures.
Surrounding Road Network	NA	NA	• Existing road infrastructure may require to be upgraded to support development. During the planning and design phases of potential upgrades, careful consideration must be given to minimising road traffic noise impacts on nearby sensitive receptors by incorporating mitigation measures such as low noise road surface, speed limits, and physical acoustic barriers where appropriate. A detailed assessment of road traffic noise should be conducted once traffic volumes for vehicles accessing and leaving the Precinct are confirmed.

Based on the noise considerations outlined in **Table 12**, acoustic consideration areas where acoustic buffers or mitigation principles are likely to be required are shown below in **Figure 3**.



Scale: 1:30,000 at A4
Coordinate System: GDA2020 MGA Zone 56

Date Drawn: 01-Jul-2026
Project Number: 630.032981.00001

Data Source: NearMap May 2026

SLR

CONSIDERATION AREAS

FIGURE 3

H:\Projects-SLR\630-SrvNTL\630-NTL\630.032981.00001 NSW DPHI-Confidential mine reh\06 SLR Data\01 GIS\GIS\630.032981_F03_SitePlan_ConsiderationAreas_A&V_04.mxd

5.0 Mitigation and Management Measures

5.1 Operational Impacts

Where operational noise impacts from development within the Precinct are predicted to exceed the relevant noise criteria, feasible and reasonable operational noise mitigation and management measures should be considered, with the aim of reducing noise emissions to the relevant criteria.

The typical hierarchy for mitigation and management of industrial noise sources is as follows:

- Land-use control
  - Separating noise producing industries from sensitive areas.
  - Provision of buffer distances between sensitive receivers and high noise generating developments
  - Manage cumulative construction and operational traffic noise associated with staged development.
- Reducing noise emissions at the source (i.e. noise source control)
  - Selecting quieter equipment
  - Enclosing the source; the design of the enclosure and materials used to absorb sound will affect the attenuation achieved
  - Silencing exhausts – muffler design and noise barrier systems
  - Active noise control, effective on a limited range of noise sources
  - Times of operation
  - Smooth roadways and vehicular access points
- Reducing noise in transmission to the receiver (i.e. noise path control)
  - Noise barriers: more effective if near source or receiver; effectiveness also controlled by materials used (reflective or absorptive) and by height
  - Mounds or bunds
  - Site design to maximise the distance from critical noise sources to the receiver and with intervening buildings to act as barriers
- Reducing noise at the receiver (i.e. at-receiver control).
  - Insulation
  - Upgrade glazing of windows and use of mechanical ventilation and/or air conditioning
  - Other mutually accepted trade-offs for benefits
  - Voluntary acquisition.

In addition to the above, development applications for noise-sensitive uses, or for uses with the potential to generate noise or vibration impacts, should be supported by a noise and vibration assessment that identifies likely impacts and appropriate mitigation measures.

5.2 Operational Impacts from Existing Rail Loop

Non-network rail lines servicing one or more industrial sites should develop a noise and vibration management plan, to be approved by the relevant planning authority. The following specific noise measures should be considered as part of this plan:

- noise control should be maximised during the design stage through route selection and physical mitigation including maximising the distance between the rail line and noise sensitive land-uses where practicable, including the use of cuttings and noise barriers where feasible and reasonable
- track design should avoid intermittent noise associated with the interaction of track and rolling stock (e.g. wheel squeal)
- rolling stock permitted to use the line should be restricted to locomotives that when they were new or substantially modified were approved to operate under an environment protection licence or a Pollution Control Approval pursuant to the former Pollution Control Act 1970
- timetabling of freight movement should minimise operation during sensitive evening and night periods, where possible
- locomotives should operate at lower speeds to reduce noise emissions
- drivers should be trained to minimise engine idling and unnecessary use of train horns as part of operating conditions.

The choice of reasonable and feasible noise-control measures depends on both the degree of mitigation required and the undesirable characteristics of the noise sources that need to be controlled. The actual measures chosen will also depend on site factors, such as the ability of the site to accommodate particular engineering measures relative to other measures and their costs and overall adverse social, economic and environmental effects.

A detailed assessment of all potential feasible and reasonable mitigation measures that could be applied to the Precinct will be undertaken as part of the detailed design when more information is known.

6.0 Conclusion

SLR Consulting has prepared a Noise and Vibration Technical Assessment (NVTA) to inform planning considerations associated with the proposed rezoning of the Macquarie Coal Complex (the Precinct) and Teralba Southgate sub-precinct. This report establishes methods for determining specific noise performance criteria for operation and construction of the Precinct and Teralba Southgate sub-precinct, allowing for the investigation and development of noise control strategies during the planning and design phases.

In-principal mitigation and management strategies have been identified in this assessment. Consideration of these strategies during the planning and design phases has the potential to minimise the level of mitigation required. Appropriate mitigation strategies should be developed through later design stages more information about specific locations and noise emissions sources of industries are determined.

A detailed assessment of noise impacts on surrounding sensitive receivers has not been undertaken as part of this assessment, as the future development layout and land uses within the Precinct and Teralba Southgate sub-precinct are yet to be confirmed. Future development of the Precinct and Teralba Southgate sub-precinct will require specific assessments to consider appropriate mitigation and management measures to ensure the acoustic amenity of the local community and individual sensitive land uses surrounding the Precinct and Teralba Southgate sub-precinct are not adversely impacted.

Appendix A Acoustic Terminology

Noise and Vibration Technical Assessment

Post Mining Land Use – Rezoning - Macquarie Coal Complex Pilot Project

Department of Planning, Housing and Infrastructure

SLR Project No.: 630.032981.00001

1 July 2026

1. Sound Level or Noise Level

The terms 'sound' and 'noise' are almost interchangeable, except that 'noise' often refers to unwanted sound.

Sound (or noise) consists of minute fluctuations in atmospheric pressure. The human ear responds to changes in sound pressure over a very wide range with the loudest sound pressure to which the human ear can respond being ten million times greater than the softest. The decibel (abbreviated as dB) scale reduces this ratio to a more manageable size by the use of logarithms.

The symbols SPL, L or LP are commonly used to represent Sound Pressure Level. The symbol LA represents A-weighted Sound Pressure Level. The standard reference unit for Sound Pressure Levels expressed in decibels is 2×10^{-5} Pa.

2. 'A' Weighted Sound Pressure Level

The overall level of a sound is usually expressed in terms of dBA, which is measured using a sound level meter with an 'A-weighting' filter. This is an electronic filter having a frequency response corresponding approximately to that of human hearing.

People's hearing is most sensitive to sounds at mid frequencies (500 Hz to 4,000 Hz), and less sensitive at lower and higher frequencies. Different sources having the same dBA level generally sound about equally loud.

A change of 1 dB or 2 dB in the level of a sound is difficult for most people to detect, whilst a 3 dB to 5 dB change corresponds to a small but noticeable change in loudness. A 10 dB change corresponds to an approximate doubling or halving in loudness. The table below lists examples of typical noise levels.

Sound Pressure Level (dBA)	Typical Source	Subjective Evaluation
130	Threshold of pain	Intolerable
120	Heavy rock concert	Extremely noisy
110	Grinding on steel	
100	Loud car horn at 3 m	Very noisy
90	Construction site with pneumatic hammering	
80	Kerbside of busy street	Loud
70	Loud radio or television	
60	Department store	Moderate to quiet
50	General Office	
40	Inside private office	Quiet to very quiet
30	Inside bedroom	
20	Recording studio	Almost silent

Other weightings (eg B, C and D) are less commonly used than A-weighting. Sound Levels measured without any weighting are referred to as 'linear', and the units are expressed as dB(lin) or dB.

3. Sound Power Level

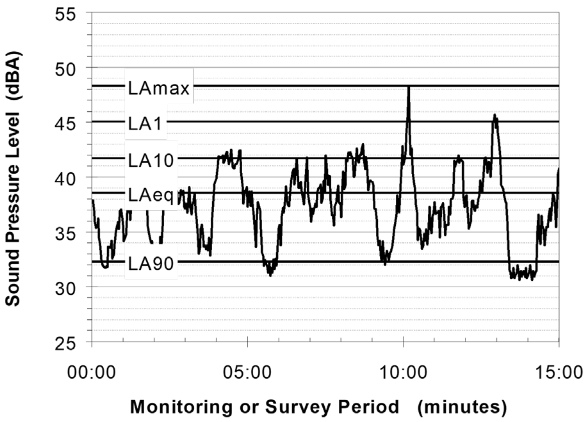
The Sound Power of a source is the rate at which it emits acoustic energy. As with Sound Pressure Levels, Sound Power Levels are expressed in decibel units (dB or dBA), but may be identified by the symbols SWL or LW, or by the reference unit 10^{-12} W.

The relationship between Sound Power and Sound Pressure is similar to the effect of an electric radiator, which is characterised by a power rating but has an effect on the surrounding environment that can be measured in terms of a different parameter, temperature.

4. Statistical Noise Levels

Sounds that vary in level over time, such as road traffic noise and most community noise, are commonly described in terms of the statistical exceedance levels LAN, where LAN is the A-weighted sound pressure level exceeded for N% of a given measurement period. For example, the LA1 is the noise level exceeded for 1% of the time, LA10 the noise exceeded for 10% of the time, and so on.

The following figure presents a hypothetical 15 minute noise survey, illustrating various common statistical indices of interest.



Of particular relevance, are:

LA1 The noise level exceeded for 1% of the 15 minute interval.

LA10 The noise level exceeded for 10% of the 15 minute interval. This is commonly referred to as the average maximum noise level.

LA90 The noise level exceeded for 90% of the sample period. This noise level is described as the average minimum background sound level (in the absence of the source under consideration), or simply the background level.

LAeq The A-weighted equivalent noise level (basically, the average noise level). It is defined as the steady sound level that contains the same amount of acoustical energy as the corresponding time-varying sound.

LAmax The A-weighted maximum sound pressure level of an event measured with a sound level meter.

5. Frequency Analysis

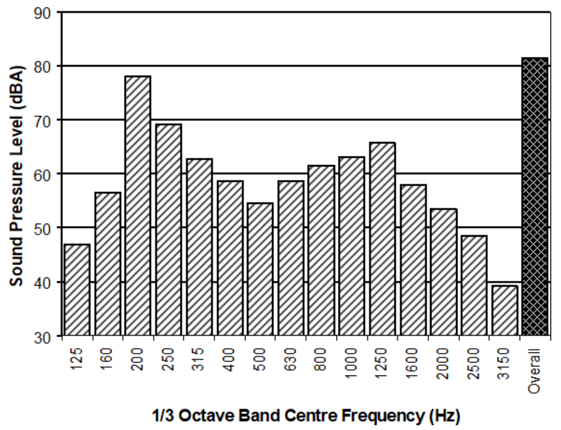
Frequency analysis is the process used to examine the tones (or frequency components) which make up the overall noise or vibration signal.

The units for frequency are Hertz (Hz), which represent the number of cycles per second.

Frequency analysis can be in:

- Octave bands (where the centre frequency and width of each band is double the previous band)
- 1/3 octave bands (three bands in each octave band)
- Narrow band (where the spectrum is divided into 400 or more bands of equal width)

The following figure shows a 1/3 octave band frequency analysis where the noise is dominated by the 200 Hz band. Note that the indicated level of each individual band is less than the overall level, which is the logarithmic sum of the bands.



6. Annoying Noise (Special Audible Characteristics)

A louder noise will generally be more annoying to nearby receivers than a quieter one. However, noise is often also found to be more annoying and result in larger impacts where the following characteristics are apparent:

- **Tonality** - tonal noise contains one or more prominent tones (ie differences in distinct frequency components between adjoining octave or 1/3 octave bands), and is normally regarded as more annoying than 'broad band' noise.
- **Impulsiveness** - an impulsive noise is characterised by one or more short sharp peaks in the time domain, such as occurs during hammering.
- **Intermittency** - intermittent noise varies in level with the change in level being clearly audible. An example would include mechanical plant cycling on and off.
- **Low Frequency Noise** - low frequency noise contains significant energy in the lower frequency bands, which are typically taken to be in the 10 to 160 Hz region.

7. Vibration

Vibration may be defined as cyclic or transient motion. This motion can be measured in terms of its displacement, velocity or acceleration. Most assessments of human response to vibration or the risk of damage to buildings use measurements of vibration velocity. These may be expressed in terms of 'peak' velocity or 'rms' velocity.

The former is the maximum instantaneous velocity, without any averaging, and is sometimes referred to as 'peak particle velocity', or PPV. The latter incorporates 'root mean squared' averaging over some defined time period.

Vibration measurements may be carried out in a single axis or alternatively as triaxial measurements (ie vertical, longitudinal and transverse).

The common units for velocity are millimetres per second (mm/s). As with noise, decibel units can also be used, in which case the reference level should always be stated. A vibration level V, expressed in mm/s can be converted to decibels by the formula 20 log (V/Vo), where Vo is the reference level (10^{-9} m/s). Care is required in this regard, as other reference levels may be used.

8. Human Perception of Vibration

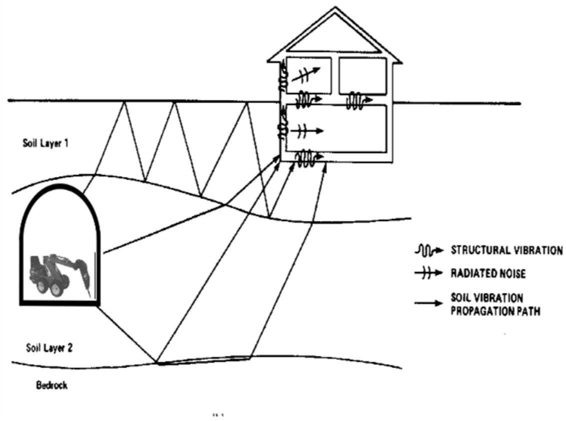
People are able to 'feel' vibration at levels lower than those required to cause even superficial damage to the most susceptible classes of building (even though they may not be disturbed by the motion). An individual's perception of motion or response to vibration depends very strongly on previous experience and expectations, and on other connotations associated with the perceived source of the vibration. For example, the vibration that a person responds to as 'normal' in a car, bus or train is considerably higher than what is perceived as 'normal' in a shop, office or dwelling.

9. Ground-borne Noise, Structure-borne Noise and Regenerated Noise

Noise that propagates through a structure as vibration and is radiated by vibrating wall and floor surfaces is termed 'structure-borne noise', 'ground-borne noise' or 'regenerated noise'. This noise originates as vibration and propagates between the source and receiver through the ground and/or building structural elements, rather than through the air.

Typical sources of ground-borne or structure-borne noise include tunnelling works, underground railways, excavation plant (eg rockbreakers), and building services plant (eg fans, compressors and generators).

The following figure presents an example of the various paths by which vibration and ground-borne noise may be transmitted between a source and receiver for construction activities occurring within a tunnel.



The term 'regenerated noise' is also used in other instances where energy is converted to noise away from the primary source. One example would be a fan blowing air through a discharge grill. The fan is the energy source and primary noise source. Additional noise may be created by the aerodynamic effect of the discharge grill in the airstream. This secondary noise is referred to as regenerated noise.

Appendix B NPfI Mitigation Measures

Noise and Vibration Technical Assessment

Post Mining Land Use – Rezoning - Macquarie Coal Complex Pilot Project

Department of Planning, Housing and Infrastructure

SLR Project No.: 630.032981.00001

1 July 2026

Best Management Practice (BMP)

Best management practice (BMP) is the application of particular operational procedures that minimise noise while retaining productive efficiency.

Where applied, these measures and practices are often documented in a noise management plan so that operational practices and undertakings are clearly understood and applied at all levels of an industrial operation. Application of BMP can include the following types of practice:

- Using the quietest plant that can do the job
- Scheduling the use of noisy equipment at the least-sensitive time of day
- Not operating, or reducing operations at night
- Siting noisy equipment behind structures that act as barriers, or at the greatest distance from the noise-sensitive area or orienting the equipment so that noise emissions are directed away from any sensitive areas, to achieve the maximum attenuation of noise
- Where there are several noisy pieces of equipment, scheduling operations so they are used separately rather than concurrently
- Keeping equipment well-maintained and operating it in a proper and efficient manner
- Using 'quiet' practices when operating equipment, for example, positioning idling trucks in appropriate areas
- Running staff-education programs and regular tool box talks on the effects of noise and the use of quiet work practices.

Best Available Technology Economically Achievable (BATEA)

With 'Best available technology economically achievable' (BATEA), equipment, plant and machinery that produce noise incorporate the most advanced and affordable technology to minimise noise output. Affordability is not necessarily determined by the price of the technology alone. Increased productivity may also result from using more advanced equipment, offsetting the initial outlay, for example, using 'quieter' equipment that can be operated over extended hours. Old or badly-designed equipment can often be a major source of noise.

Where BMP fails to achieve the required noise reduction by itself, the BATEA approach should then be considered. Examples of uses of BATEA include:

- Considering alternatives to tonal reversing alarms (where work health and safety is appropriately considered)
- Using equipment with efficient muffler design
- Using quieter engines, such as electric instead of internal combustion
- Fitting and maintaining noise reduction packages on plant and equipment
- Using efficient enclosures for noise sources
- Damping or lining metal trays or bins
- Active noise control.

For many industries there are a wide range of factors that can restrict the feasibility and reasonableness of applying BMP or BATEA measures on a particular site. Work health and safety considerations must also be taken into account as well as any other regulatory and process requirements.

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